SECURITY ADVISORY

Cisco Secure Firewall Management Center (FMC)

CVE-2026-20079
CVE-2026-20131



Memento Mori Cyber Threat Intelligence

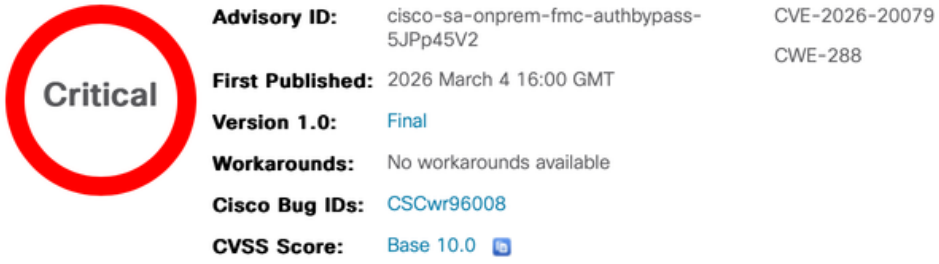
Executive Summary

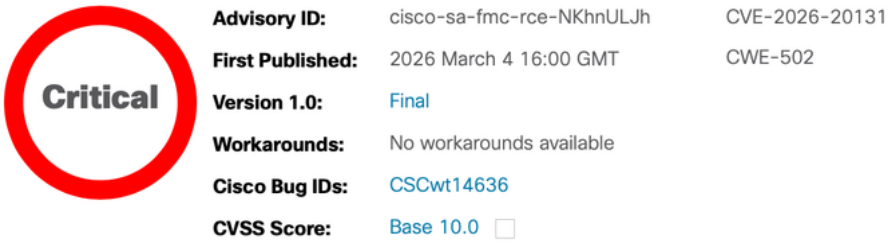
Cisco has recently disclosed multiple critical vulnerabilities affecting Cisco Secure Firewall Management Center (FMC). These vulnerabilities have been assigned the maximum severity level and may allow an attacker to gain root-level access to affected systems.

Cisco Secure FMC is a centralized management platform used to control and monitor Cisco firewall environments, including Firepower Threat Defense (FTD) devices. Because FMC acts as the central command and management node for network security infrastructure, a successful compromise could allow attackers to manipulate firewall policies, disable security protections, and potentially gain deeper access into enterprise networks.

The vulnerabilities are particularly dangerous in environments where external authentication mechanisms such as RADIUS are enabled for administrative access. In certain configurations, attackers may exploit insufficient input validation to execute arbitrary commands on the underlying operating system.

Organizations using Cisco Secure FMC should treat this advisory as high priority and immediately apply the recommended security updates.





Technical Overview

Cisco Secure Firewall Management Center is designed to centrally manage multiple security components across an enterprise network, including:

- Firewall rule configuration
- Intrusion Prevention System (IPS) policies
- URL filtering
- Malware detection
- Device monitoring and logging

Because FMC acts as a management plane for security infrastructure, vulnerabilities within this system pose significant operational risks.

The recently disclosed vulnerabilities stem from improper handling of authentication data and insufficient input validation within specific FMC services. Under certain circumstances, maliciously crafted authentication requests can trigger unintended command execution on the system.

The vulnerabilities allow attackers to inject commands during authentication processing, which may result in execution of arbitrary code with elevated privileges.
Once exploited, attackers may obtain root-level control over the FMC appliance.

Possible Attack Scenario

A realistic attack scenario may unfold as follows:

1. An attacker identifies an exposed Cisco FMC management interface.
2. The attacker detects that RADIUS authentication is enabled for administrative access.
3. The attacker sends a specially crafted authentication request containing malicious input.
4. The vulnerability allows the injected commands to be executed on the underlying system.
5. The attacker gains root privileges on the FMC appliance.
6. The attacker may then modify firewall policies, disable security monitoring, or pivot into managed firewall devices.
7. Because FMC centrally manages multiple firewalls, a single compromise may allow attackers to impact multiple security devices simultaneously.

Affected Environments

- Cisco Secure Firewall Management Center
- Cisco Firepower Threat Defense environments managed via FMC
- Administrative access through RADIUS authentication

Environments where management interfaces are exposed to untrusted networks face an increased risk of exploitation.

Security teams should carefully review whether FMC management interfaces are accessible from external networks or broadly accessible internal segments.

Mitigation and Recommended Actions

Organizations using Cisco Secure FMC should implement the following mitigation measures immediately.

- Apply Security Updates
- Cisco has released software updates addressing the vulnerabilities.

All affected systems should be updated to the latest secure versions as soon as possible.

Restrict Management Access

Administrative interfaces should never be exposed to public networks.

- Restrict FMC management access to trusted administrative networks
- Implement network segmentation for security infrastructure
- Limit administrative access through VPN or secure management networks